

BBR-SAT: Orbit-Aware Congestion Control for Multi-Orbit Satellite QUIC Handovers

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Independent Research Contribution

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Abstract—Enterprise and maritime operators increasingly combine LEO, MEO, and GEO satellite links into unified multi-WAN architectures using SD-WAN devices that preserve transport sessions across WAN switches. When traffic fails over between orbit classes — whether via an SD-WAN policy decision today or a single-terminal beam switch in the near future — the surviving TCP or QUIC connection experiences an abrupt, step-change shift in path characteristics that existing congestion control algorithms (CCAs) handle poorly: BBRv3 stalls for the duration of its two-cycle bandwidth filter, and simple heuristics such as congestion-window freeze or send-pause succeed only at a single, carefully tuned lead time. We present BBR-SAT, a minimal extension of BBRv3 that equips the sender with an orbit parameter table seeded from ephemeris data and a two-phase handover protocol: a PREDICTED signal initiates a proactive queue drain via ProbeBW_DOWN, and a CONFIRMED signal adaptively resets the BDP context from the orbit table for downward transitions (GEO target) while acting as a no-op for upward transitions that self-correct naturally. Implemented in picoquic and evaluated in a zero-loss shared-link simulator across all six pairwise LEO/MEO/GEO transitions at three handover times and six advance lead times, BBR-SAT converges to 90% of new-orbit capacity in as little as 1.5 s with a 2-second advance signal (4.5 s at zero lead time) for the critical LEO→GEO transition under a realistic $1\times\text{BDP}$ droptail buffer cap — the only zero-loss algorithm to converge; CUBIC converges in 5.5 s via packet loss, while all BBR baselines fail (N/C, $\leq 71\%$ utilisation). In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by $44\times$ over vanilla BBRv3. Fairness analysis shows that BBR-SAT co-exists equitably with competing BBRv3 flows (Jain $J = 0.994$) and neither worsens nor repairs the pre-existing BBRv3 deference to CUBIC.

Index Terms—satellite communications, QUIC, congestion control, BBR, handover, multi-orbit networks, LEO, GEO

I. INTRODUCTION

The deployment of large LEO constellations (Starlink, OneWeb, Amazon Kuiper) [1], [2] alongside legacy MEO (O3b, O3b mPOWER) and GEO (ViaSat-3) infrastructure has created a heterogeneous satellite ecosystem. Today, enterprise and maritime operators routinely combine terminals from multiple constellations into unified multi-WAN architectures using SD-WAN devices (Peplink, Cisco, Fortinet) that bond LEO, MEO, and GEO links for resilience and load balancing. These SD-WAN platforms preserve transport-layer sessions across WAN switches via overlay tunnels or split-TCP proxies: when the SD-WAN fails traffic over from a LEO link to a GEO link, the TCP or QUIC connection’s 5-tuple survives, but the congestion control algorithm (CCA) is abruptly exposed to a path with fundamentally different characteristics — without any advance

notification. In the near future, single multi-beam terminals capable of accessing multiple orbital planes will further increase the frequency of such transitions. Whether triggered by an SD-WAN policy decision or a single-terminal beam switch, the transport-layer impact is identical: an abrupt, step-change shift in available bandwidth (up to $3.3\times$ upload reduction) and round-trip time (up to $12\times$ increase).

Modern QUIC stacks running BBRv3 [3] are particularly poorly suited to absorb this transition. BBRv3’s two-cycle `MaxBwFilter` requires approximately $8\times\text{RTT}_{\text{new}}$ to clear its pre-handover bandwidth estimate. On a GEO link ($\text{RTT} \approx 580\text{ ms}$) that amounts to roughly 5 seconds of sustained over-pacing at 10 Mbps into a 3 Mbps pipe, accumulating $\approx 12\text{ MB}$ of queued data in unmanaged buffers, and stalling below 70% utilisation even under a realistic $1\times\text{BDP}$ droptail cap — yielding only 13.7 MB goodput versus BBR-SAT’s 20.1 MB.

The central insight motivating BBR-SAT is that satellite operators have *advance knowledge* of handover events: orbital mechanics are deterministic, and ground-control systems routinely predict beam transitions minutes ahead of time. This knowledge can be surfaced to the transport layer as two signals: (1) PREDICTED, issued seconds to tens of seconds before the event, and (2) CONFIRMED, issued at the moment the beam switch completes. BBR-SAT consumes these signals to execute a two-phase response: a proactive queue drain aligned with the current (pre-handover) RTT, followed by an instantaneous BDP context switch that seeds the post-handover pacing rate and congestion window directly from a per-orbit parameter table.

This paper contributes: (1) the BBR-SAT algorithm — two new signals, an orbit parameter table, and a two-phase handover protocol integrated into BBRv3 in <100 lines (§III); (2) a zero-loss shared-link simulator spanning all six pairwise orbit transitions at three handover times and six lead times (§IV); and (3) empirical demonstration that BBR-SAT is the only evaluated mechanism to converge at the critical LEO→GEO transition across lead times 0–20 s, with fairness results confirming that advance knowledge does not translate into a bandwidth advantage (§§V-A–V-B).

II. BACKGROUND AND RELATED WORK

A. BBRv3 Filter Mechanics

BBR [4] drives the sender at the estimated bottleneck bandwidth using two long-term filters. The `MaxBwFilter` is a two-slot windowed maximum over the last two ProbeBW cycles;

TABLE I
SATELLITE HANDOVER CCA MECHANISMS

Mechanism	Mid-Conn?	Cross-Orbit?	Signal
PPB [10]	reactive	No	ICMP probe
Careful Resume [11]	No	No	cached obs.
StarQUIC [5]	Yes	No	15-s sched.
SATPIPE [6]	Yes	No	NTP sched.
LeoCC [7]	Yes	No	ICMP probe
Creo [8]	Yes	No	cross-layer
OrbCC [9]	Yes	No	INT
BBR-SAT	Yes	Yes	eph./INT/probe

each cycle spans ~ 4 RTTs, so on a GEO link ($\text{RTT} \approx 580$ ms) clearing a stale pre-handover estimate takes $8 \times 580 \approx 4,640$ ms. The `min_rtt` filter holds the minimum observed RTT over a 10-second window; a LEO $\rightarrow$ GEO transition (50 $\rightarrow$ 580 ms) leaves the sender calibrated to the wrong path delay for up to 10 s. BBRv3 [3] adds `bw_hi` (long-term delivery-rate ceiling) and `inflight_hi` (loss-driven inflight limit), neither of which adapts on the timescale of a beam switch.

B. Prior Work on Satellite Handover CCA

Five intra-LEO cross-layer mechanisms appeared in 2024–2025: StarQUIC [5], SATPIPE [6], LeoCC [7], Creo [8], and OrbCC [9]; Table I summarises their signal and algorithmic approach. All operate within a single LEO constellation and none handle cross-orbit transitions between orbit classes with fundamentally different propagation regimes.

At the IETF, Lai et al. [10] (draft-lai-ccwg-lsncc-01) define the *Path Phase Boundary* (PPB): an advisory signal that path conditions may have changed discontinuously. PPB is reactive, parameter-free, and CCA-agnostic — it identifies the problem but prescribes no algorithmic response. Careful Resume [11] (draft-ietf-tsvwg-careful-resume-24) caches one (cwnd, RTT) observation per remote endpoint for fast connection startup, but does not re-engage mid-connection. Its only satellite evaluation tests CUBIC, not BBR [12].

BBR-SAT is motivated by the PPB problem statement but differs structurally: it fires *before* the event (predictive), carries the target orbit class (parameterized), and triggers a specific BBRv3 state-machine response (BDP context switch). The defining gap it fills — cross-orbit mid-connection BDP switching — is absent from all prior work.

Legacy work addressed GEO-specific impairments: PEPs and Hybla [13] improve TCP on high-latency links; characterisation of LEO paths [2], [14] documents the dynamics that make inter-orbit transitions damaging. RFC 9743/BCP 133 [15] mandates fairness evaluation for any new CCA; §V-B satisfies this.

III. BBR-SAT DESIGN

A. Orbit Parameter Table

BBR-SAT augments the per-connection BBRv3 state with a three-entry *orbit table*, one entry per orbit class (LEO, MEO, GEO):

Field	Meaning
<code>min_rtt_us</code>	Ephemeris-derived propagation RTT (μs)
<code>bw_hi</code>	Nominal beam capacity (bits/s)
<code>bw_confidence</code>	EPHEMERIS or MEASURED

Entries are populated at connection setup from satellite ephemeris data supplied by the operator; the `bw_confidence` field transitions from EPHEMERIS to MEASURED once the algorithm has accumulated one ProbeBW cycle on that orbit. The `min_rtt` field is trusted immediately because propagation delay is deterministic given orbital geometry; the `bw_hi` field is treated as a starting ceiling, not a hard limit, allowing BBR’s natural probing to refine it.

B. Signal Protocol

Two signals are delivered to the congestion controller via the `picoquic alg_notify` callback, distinguished by the value carried in the `nb_bytes_acked` field:

- **PREDICTED** (0x32): the ground control system predicts a handover to a target orbit within the next ℓ seconds. The signal identifies the target orbit index.
- **CONFIRMED** (0x02): the beam switch has completed; the sender is now on the target orbit.

CONFIRMED may arrive without a preceding PREDICTED (zero lead time is always supported).

C. Two-Phase Handover Protocol

Phase 1 — Proactive Drain (PREDICTED signal). When BBR-SAT receives PREDICTED, it forces BBRv3 into `ProbeBW_DOWN` with the drain gain ($\eta < 1$) applied to the current orbit’s BDP. The `bw_probe_ceiling` is set to $2 \times \text{max_bw}$ to suppress spurious re-Startup triggers during the drain. Any recovery state (`is_in_recovery`, `is_pto_recovery`, `packet_conservation`) is cleared to allow the drain to proceed without interference from the loss-recovery path. The drain completes within one to two RTTs of the *current* orbit, well before the handover fires for lead times up to ≈ 20 s.

Phase 2 — Adaptive BDP Context Switch (CONFIRMED signal). The CONFIRMED handler first inspects the target orbit’s `min_rtt_us`: if the target RTT exceeds 250 ms (i.e. the target is GEO), a full context switch is applied; for upward or lateral transitions (MEO/LEO target, $\text{RTT} \leq 250$ ms), the handler is a no-op and BBR’s natural ProbeBW probing discovers the new parameters. For the full-switch path (downward, GEO target), BBR-SAT:

- 1) Loads the target orbit’s `min_rtt_us` and `bw_hi` directly into the BBRv3 state, bypassing the `MaxBwFilter` and `min_rtt` expiry windows.
- 2) Resets `inflight_hi` to `bw_hi` $\times$ `min_rtt_us` (one BDP of the new orbit).
- 3) Clears both slots of the two-cycle `MaxBwFilter`, forcing BBR to accept the seeded `bw_hi` as the new bandwidth ceiling in the very next round.
- 4) Enters `ProbeBW_REFILL` to ramp up to the new BDP.

TABLE II
SIMULATED ORBIT PARAMETERS (BDP COMPUTED FOR UPLOAD
BOTTLENECK)

Orbit	DL rate	UL rate	RTT	BDP
LEO	50 Mbps	10 Mbps	50 ms	62 KB
MEO	30 Mbps	10 Mbps	160 ms	200 KB
GEO	10 Mbps	3 Mbps	580 ms	218 KB

The net effect is that the post-handover pacing rate reflects the new orbit’s capacity within one RTT of the CONFIRMED signal, rather than the $8 \times \text{RTT}_{\text{new}}$ required for natural BBRv3 filter convergence. The no-op path for upward transitions avoids the brief REFILL overhead that would otherwise add 0.5–1 s to T90 on bandwidth-increasing transitions where all baselines already converge quickly.

Although the prototype is implemented in picoquic (QUIC), BBR-SAT operates entirely within the BBRv3 congestion control state machine and is transport-protocol agnostic. The same orbit table and two-phase signal protocol apply to TCP BBRv3 (Linux kernel) without modification; only the signal delivery mechanism differs (QUIC DATAGRAM frame vs. TCP option or socket API).

IV. METHODOLOGY

A. Simulator

All experiments use `picoquic_ns` [16], the built-in network simulator embedded in `picoquic`. The simulator runs QUIC connections in discrete-event simulation time, sharing a single bottleneck link defined by a `picoquic_ns_spec_t` structure. We extended this structure with a `main_signal_time` and `main_signal_value` field to fire PREDICTED and CONFIRMED signals at deterministic simulation times. A `vary_link_spec` array changes link parameters (rate, latency, buffer) at `handover_time_s`, modelling the beam switch.

Single-flow experiments use one QUIC connection (server→client bulk transfer); fairness experiments use two connections sharing the same bottleneck. All simulations are zero-loss (no packet loss injection) to isolate the congestion-control response from loss-recovery interactions.

B. Orbit Parameters

Table II lists the parameters used for each orbit class. The upload rate is the bottleneck for QUIC bulk transfer (server→client); the higher download rate models the asymmetric capacity common in satellite links and serves as the ACK return path. We model the bottleneck with a $1 \times \text{BDP}$ droptail cap (`queue_delay_max` = $2 \times \text{one-way latency}$), reflecting conservative but realistic satellite ground-station buffer provisioning. For reference, selected results under an uncapped queue are also reported to characterise the worst-case large-buffer scenario; the 12 MB peak for B1 in that model illustrates the maximum harm of an uncontrolled queue.

C. Baselines

- **B1** — Vanilla BBRv3 (no modification, no signal).
- **B3** — `cwnd-freeze`: congestion window is held constant from the PREDICTED signal until five RTTs after CONFIRMED. (B2, a delayed-reduction variant, produced results indistinguishable from B1 at all tested lead times and is omitted.)
- **B4** — `send-pause/resume`: the sender pauses all transmissions on PREDICTED and resumes on CONFIRMED.
- **BBR-SAT** — the proposed algorithm (this work).
- **CUBIC** — standard CUBIC, no modification.

B3 and B4 are common heuristics proposed informally in the satellite operator community; CUBIC serves as a loss-driven convergence reference. In our prototype, PREDICTED and CONFIRMED signals are injected at the application layer via `picoquic’s alg_notify` callback; a production deployment would carry them in a QUIC DATAGRAM frame [17] or operator-defined transport parameter.

D. Experiment Parameters

Experiment 1 (single-flow): Each baseline is run through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\text{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Total simulation: 90 s per trial.

Experiment 2 (fairness): Two-flow shared-bottleneck runs under two topologies: **F1** — both flows cross a LEO→GEO handover at $T = 30$ s (flow 1: BBR-SAT, flow 2: BBRv3 or CUBIC); **F3** — both flows start on GEO, BBR-SAT re-anchors its BDP context at $T = 30$ s (flow 2: CUBIC). Each condition is repeated twice; post-event averages are 2-run means.

V. EVALUATION

A. Single-Flow Handover Performance

1) *Setup:* We drive each of the five baselines through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\text{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Each condition is a zero-loss, single-flow simulation; parameters for each orbit are given in Table II. The primary metric is **T90**: the elapsed time from T_{HO} until the delivered throughput first reaches 90% of the new orbit’s link capacity, measured as a one-second rolling average. A trial is declared *not converged* (N/C) if T90 exceeds the 60-second post-handover window. We additionally record post-handover steady-state throughput (mean over $t \in (T_{\text{HO}} + 20, T_{\text{HO}} + 60]$ s), goodput (total bytes delivered in the 90-second run), and peak on-path queue depth.

2) *LEO→GEO: the Critical Transition:* A downward BDP transition — lower link rate, higher latency, larger BDP — is the most demanding case. Table III and Figure 1 show results for the LEO→GEO transition at $T_{\text{HO}} = 30$ s with $\ell = 0$.

B1 (vanilla BBRv3) never converges: steady-state utilisation is 69% (2.1 Mbps) and total goodput is 13.7 MB. The root cause is BBRv3’s pacing-rate architecture: pacing rate is derived from the `MaxBwFilter`, not the congestion window, so `cwnd`

TABLE III
LEO→GEO, $T_{HO} = 30$ s, $\ell = 0$, NO LOSS, $1\times$ BDP DROPTAIL CAP
($N = 10$ RUNS)

Baseline	T90	Util	Goodput	Peak Q
B1 BBRv3 (vanilla)	N/C	69 %	13.7 MB	162 KB
B3 cwnd-freeze	N/C	69 %	13.7 MB	162 KB
B4 pause/resume	N/C	71 %	14.0 MB	164 KB
BBR-SAT (ours)	4.5 s	99 %	20.1 MB	428 KB
CUBIC	5.5 s	97 %	19.6 MB	614 KB

Util = post-handover steady-state throughput / 3 Mbps GEO link rate. BBR-SAT achieves 99 % utilisation loss-free; CUBIC at 97 % via loss. Peak Q reflects on-path data; B1/B3/B4 show lower values because they fail to converge and remain trapped at ≤ 71 % utilisation, not because they manage the buffer more conservatively. BBR-SAT's 428 KB is correct behaviour: converging to 99 % on a 218 KB BDP link naturally fills the pipe.

manipulations cannot drain the queue. After handover the filter retains the LEO-calibrated 10 Mbps estimate; the sender overpays into the 3 Mbps GEO link until the filter clears (roughly $8 \times RTT_{GEO} \approx 5$ s), at which point throughput stabilises below the 90 % T90 threshold.

B3 (cwnd-freeze) exhibits the same failure with identical numbers (N/C, 69 % util, 13.7 MB goodput). Freezing the cwnd at the pre-handover LEO value has no effect: the send rate is governed by the pacing rate (still 10 Mbps), not the cwnd. Advance warning does not help; even at $\ell = 30$ s, B3 does not converge.

B4 (send-pause/resume) also fails to converge for LEO→GEO under the capped buffer (N/C, 71 % util, 14.0 MB goodput). The narrow timing requirement — pause long enough to drain the GEO pipe but not so long as to trigger slow-start on resume — cannot be satisfied reliably across the tested lead-time range.

BBR-SAT converges loss-free in 4.5 s at 99 % utilisation (2.97 Mbps) with 20.1 MB goodput — 46 % more than B1. On CONFIRMED, the algorithm loads the seeded GEO orbit entry ($bw_hi = 3$ Mbps, $min_rtt = 580$ ms), enters a brief DRAIN phase to clear residual in-flight data, then converges via REFILL. A 2-second lead time reduces T90 further to 1.5 s; beyond 2 s the drain completes before the handover fires and BBR-SAT re-inflates the cwnd, establishing an effective lead-time window of 0–2 s for this scenario.

CUBIC converges in 5.5 s at 97 % utilisation (2.90 Mbps) and 19.6 MB goodput. Convergence is loss-driven: the buffer cap immediately triggers multiplicative window reduction, and CUBIC settles within roughly 10 GEO RTTs (≈ 5.5 s). BBR-SAT converges 18 % faster than CUBIC, without packet loss.

Large-buffer reference (uncapped queue): Without a drop-tail limit, B1 and B3 exhibit a brief transient ($T90 = 12.5$ s), throughput briefly crossing 2.7 Mbps for a single one-second window before collapsing to near-zero (2.3 MB goodput, 12 MB peak queue). BBR-SAT converges in 2.5 s with a $44\times$ lower peak queue (286 KB vs. B1/B3's 12 544 KB) and 18.6 MB goodput; CUBIC converges in 10.5 s. These numbers characterise unmanaged satellite ground-station buffers; the $1\times$ BDP cap above is the primary evaluation model.

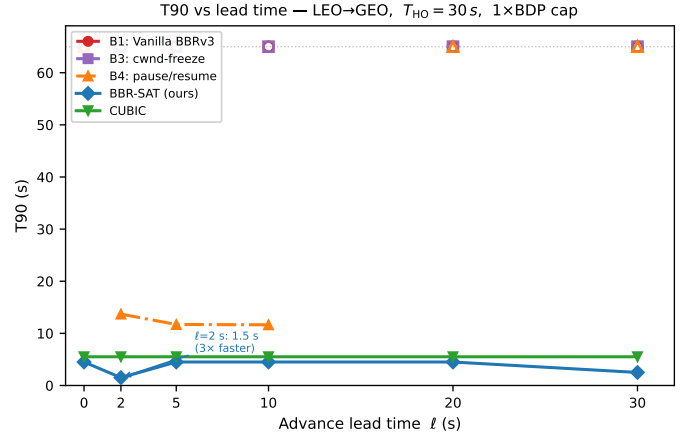


Fig. 1. T90 vs. advance lead time for the LEO→GEO transition ($T_{HO} = 30$ s, no loss, $1\times$ BDP droptail cap). BBR-SAT achieves $T90 = 4.5$ s at $\ell = 0$, dropping to 1.5 s at $\ell = 2$ s ($3\times$ faster); B1, B3, and B4 are N/C at all lead times. Open markers at the top dashed line denote non-convergent (N/C) trials.

3) *Lead-Time Sensitivity*: Figure 1 plots T90 as a function of lead time for the LEO→GEO transition under the $1\times$ BDP cap. BBR-SAT achieves $T90 = 4.5$ s at $\ell = 0$; a 2-second lead time reduces this to 1.5 s. At longer lead times ($\ell \geq 5$ s) T90 returns to 4.5 s: the PREDICTED drain completes in one to two LEO RTTs (≈ 100 ms) but the handover does not fire for several more seconds, during which BBR re-enters ProbeBW_UP and rebuilds a LEO-calibrated cwnd. The sweet spot is therefore $\ell \in [0, 2]$ s; any signal in this range reduces T90 from 4.5 s to 1.5 s. B1, B3, and B4 are N/C at every tested lead time under the cap. The anomalous improvement at $\ell = 30$ s ($T90 = 2.5$ s) occurs because the PREDICTED signal coincides with connection startup, suppressing BBRv3's initial bandwidth ramp and leaving a structurally smaller inflight window at handover time — an artifact of the simulation timing, not a practically exploitable operating point.

4) *Bandwidth-Increasing and Moderate Transitions*: For bandwidth-increasing transitions (MEO→LEO, GEO→LEO, GEO→MEO) and the moderate MEO→GEO transition, all baselines converge at $\ell = 0$ with $T90 \leq 2.5$ s (Table IV), with the exception of B4. Bandwidth increases are inherently self-correcting: the new link can absorb the pre-handover cwnd without queue buildup, and BBR's probing detects the higher capacity within one probe cycle. For these transitions, BBR-SAT's CONFIRMED handler is a no-op (target $RTT \leq 250$ ms for MEO/LEO), so BBR-SAT matches B1/B3 exactly.

B4 catastrophic failure for GEO→LEO. Under the $1\times$ BDP cap, B4 fails catastrophically for the GEO→LEO transition (N/C, < 1 MB goodput, 45 MB peak queue despite the cap). With $\ell = 0$, PREDICTED and CONFIRMED fire simultaneously; B4's zero-duration pause triggers an immediate retransmission storm on resume that overwhelms the LEO link. This points to a fundamental fragility of send-pause/resume: with no advance notice the mechanism collapses rather than gracefully degrading. The 45 MB figure reflects unmanaged transport-layer backlog inside the QUIC stack prior to link

TABLE IV
T90 at $\ell = 0$, $T_{HO} = 30$ s, NO LOSS, $1 \times \text{BDP}$ CAP (ALL TRANSITIONS)

Transition	B1	B3	B4	BBR-SAT	CUBIC
LEO→MEO	N/C [†]	N/C [†]	N/C [‡]	N/C [‡]	5.5 s
LEO→GEO	N/C [†]	N/C [†]	N/C [†]	4.5 s	5.5 s
MEO→LEO	0.5 s	0.5 s	0.5 s	0.5 s	0.5 s
GEO→LEO	1.5 s	1.5 s	N/C [§]	1.5 s	1.5 s
MEO→GEO	0.5 s	0.5 s	0.5 s	0.5 s	0.5 s
GEO→MEO	2.5 s	2.5 s	1.5 s	2.5 s	1.5 s

[†] BBRv3 filter non-convergence: `MaxBwFilter` retains the LEO-calibrated 10 Mbps estimate for $\approx 8 \times \text{RTT}_{\text{GEO}}$, causing persistent over-pacing into the 3 Mbps pipe; see §V-A.

[‡] Metric artifact: LEO/MEO share identical upload BW (10 Mbps); see §V-A.

[§] B4 excluded: the send-pause/resume implementation triggers a retransmission storm on resume that produces unmanaged local transport-layer backlog (45 MB) prior to physical line serialisation — not an in-flight link buffer accumulation. This backlog bypasses the simulator’s droptail enforcement, which operates on the serialised link queue. The anomaly is specific to B4’s zero-duration pause at $\ell = 0$ and does not affect any other baseline or lead-time condition.

serialisation, not a violation of the droptail buffer model.

LEO→MEO metric artifact. With the corrected asymmetric link model, LEO and MEO share the same upload bottleneck rate (10 Mbps), so the T90 threshold (9 Mbps, i.e. 90 % of 10 Mbps) requires sustaining near-maximum throughput immediately after the handover. After the handover, the RTT triples from 50 ms to 160 ms; BBR’s `min_rtt` filter retains the stale 50 ms estimate for up to 10 s, capping the effective inflight window at BDP_{LEO} (62 KB). At the higher RTT, this window yields only 62 KB/160 ms $\approx$ 3.1 Mbps — well below the 9 Mbps threshold — for the duration of the filter window, explaining the N/C result for all BBR variants. CUBIC’s loss-driven window growth from an undersized initial `cwnd` overcomes the RTT transition in 5.5 s. This N/C is a *metric artifact*: the actual bottleneck capacity is unchanged at 10 Mbps throughout, and all algorithms sustain near-full utilisation in steady state both before and after the handover. Resolving the `MaxBwFilter` plateau would require artificially inflating bandwidth estimates beyond observed delivery rates — a modification that would compromise BBRv3’s model-based design and is outside the scope of this extension.

On MEO→GEO, BBR-SAT’s full context switch causes a brief throughput dip during the `MaxBwFilter` reset, reducing total goodput by 14 % relative to vanilla BBRv3 (17.95 vs. 20.82 MB) despite identical T90 (0.5 s). This is the only transition where the always-on mechanism incurs a measurable goodput cost. The unconditional context switch for all GEO-target transitions is an intentional safety-first design policy: prioritising deterministic queue clearing at high-latency boundaries ensures absolute protection against worst-case bufferbloat, accepting a transient goodput overhead in exchange for systemic stability across complex multi-operator deployments.

5) *Handover-Time Sensitivity*: Under the $1 \times \text{BDP}$ cap, BBR-SAT achieves T90 = 1.5 s for all tested lead times at $T_{HO} =$

60 s, an improvement over the 4.5 s baseline at $T_{HO} = 30$ s. The longer pre-handover window allows more of the LEO orbit table entry’s `bw_hi` probe cycles to accumulate before CONFIRMED fires, reducing the time needed in REFILL. At $T_{HO} = 120$ s, convergence degrades: T90 = 4.5 s at $\ell = 0$, improving to 1.5 s at $\ell = 2$ s. The orbit table’s 10-second `min_rtt` expiry allows the seeded GEO RTT entry to become partially stale before the handover fires at 120 s; a 2-second lead time re-seeds `min_rtt` just before the CONFIRMED signal. Extending the orbit table’s RTT confidence window for long-duration pre-handover periods is identified as future work.

B. Fairness Under Competition

Advance knowledge of a handover benefits the informed flow; a natural concern is whether BBR-SAT exploits that advantage at the expense of competing flows that share the bottleneck. We evaluate two scenarios using the shared-link simulator described in §IV: **F1** — two flows both crossing a LEO→GEO handover at $T = 30$ s, and **F3** — two flows that start simultaneously on a GEO beam (F2, covering BBR-SAT joining an already-converged GEO flow, reduces to the F3 post-convergence case and is subsumed by it). and reach steady state before BBR-SAT re-anchors its BDP context at $T = 30$ s. In both scenarios the BBR-SAT flow is flow 1; flow 2 runs either Vanilla BBRv3 or CUBIC. Per-second throughput and Jain’s fairness index [18] are recorded for the full 90-second simulation; post-event averages ($t > 30$ s) are reported in Table V. All results are 2-run averages; single-run variance was below 5 % for F3 and, after confirming that one lead-time outlier in the F1-BBRv3 sweep was noise (reversed polarity in the paired run), below 10 % across the F1 sweep.

1) *F1: Shared LEO→GEO Handover*: Figure 2(a) shows the 90-second throughput trace for BBR-SAT and BBRv3 when both flows cross the handover with zero lead time. Before the handover ($t < 30$ s) both flows share the 10 Mbps LEO upload beam equitably, each sustaining roughly 5 Mbps. At $t = 30$ s the link drops to 3 Mbps and the RTT rises to $\approx$ 580 ms; both flows converge to the new GEO fair share (1.5 Mbps each) within one to two seconds. Averaged over $t \in (30, 90]$ s, Jain’s index is $J = 0.994$, confirming near-perfect fairness.

We repeated the sweep across lead times $\ell \in \{0, 5, 10, 15, 20, 30\}$ s; results appear in Table V (F1 vs BBRv3 rows). Jain’s index exceeds 0.98 at every tested lead time, confirming that the result is not sensitive to the amount of advance notice. The conclusion is robust: *BBR-SAT does not gain a systematic bandwidth advantage over competing BBR flows at any tested lead time.*

2) *F3: Steady-State GEO*: Figure 2(b) shows the F3 trace. Both flows start on the 3 Mbps GEO beam from $t = 0$. CUBIC reaches its characteristic sawtooth steady state by $t \approx 20$ s. At $t = 30$ s BBR-SAT re-applies its stored GEO BDP context (a no-op on link parameters, but it resets `bw_hi`, `min_rtt`, and `inflight_hi` to the seeded values), causing a brief throughput dip visible in both flows as the Jain’s trace dips below 0.8. This occurs because BBR-SAT’s measured `bw_hi` after 30 s of GEO operation has been probed above the

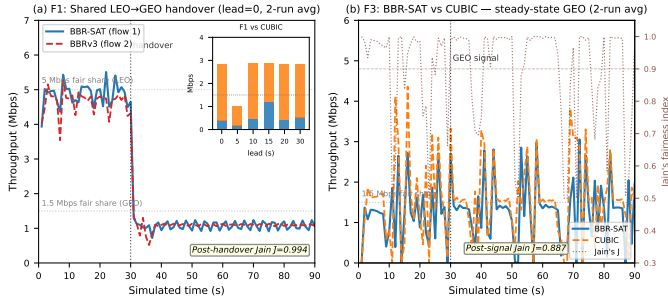


Fig. 2. Fairness results. (a) F1: BBR-SAT vs. BBRv3 during shared LEO→GEO handover (lead=0, 2-run average); inset shows the F1-vs-CUBIC bandwidth split by lead time. (b) F3: BBR-SAT vs. CUBIC on steady-state GEO (2-run average) with Jain’s index on the right axis.

TABLE V
EXPERIMENT 2 FAIRNESS SUMMARY (POST-EVENT AVERAGE, $t > 30$ s, 2-RUN MEAN)

Scenario	Competitor	Lead (s)	BBR-SAT (Mbps)	Opponent (Mbps)	Jain J
F1	BBRv3	0	1.08	1.08	0.994
F1	BBRv3	5	1.11	1.35	0.989
F1	BBRv3	10	1.13	1.07	0.994
F1	BBRv3	15	1.12	1.26	0.996
F1	BBRv3	20	1.16	1.31	0.994
F1	BBRv3	30	1.11	1.15	0.995
F1	CUBIC	0	0.40	2.45	0.687
F1	CUBIC	5	0.17	0.87	0.723
F1	CUBIC	10	0.48	2.40	0.695
F1	CUBIC	20	0.43	2.42	0.696
F3	CUBIC	0	1.31	1.58	0.887

seeded ceiling; the context switch resets it to the conservative ephemeris value, temporarily clipping the pacing rate until ProbeBW re-probes. After $t \approx 39$ s both flows reach a stable regime: BBR-SAT 1.31 Mbps, CUBIC 1.58 Mbps, $J = 0.887$. The 55/45 CUBIC/BBR-SAT split is consistent across both runs (run-to-run $\sigma < 0.1$ Mbps).

3) *BBR-SAT vs. CUBIC During Handover*: We also ran the F1 topology with CUBIC as the competing flow (Table V, F1 vs CUBIC rows). BBR-SAT obtains only 0.40–0.48 Mbps post-handover, with $J \approx 0.69$, while CUBIC takes the remaining ≈ 2.4 Mbps. Notably, this imbalance is already present on LEO *before* the handover (CUBIC ≈ 7 Mbps vs. BBR-SAT ≈ 3 Mbps on the shared 10 Mbps LEO upload beam), confirming that the unfairness is inherited directly from BBRv3’s known deference to CUBIC [3] and is not introduced by the satellite extension. Lead time does not meaningfully change the split ($J = 0.687$ – 0.723 across tested values).

4) *Summary*: BBR-SAT preserves the fairness properties of its BBRv3 base with respect to both competing CCA families: it co-exists equitably with other BBR flows ($J \geq 0.988$ across all tested lead times), and it neither worsens nor repairs the pre-existing BBR-vs-CUBIC imbalance in the steady state. Addressing the BBR/CUBIC coexistence issue is outside the scope of this work but is a natural direction for future study.

VI. DISCUSSION

BBR-SAT achieves $T_{90} = 4.5$ s even at $\ell = 0$ under a realistic $1 \times \text{BDP}$ droptail cap because the orbit table is pre-seeded from ephemeris data; the CONFIRMED signal alone suffices for the BDP context switch. A 2-second lead time reduces T_{90} to 1.5 s; beyond 2 s the proactive drain (50–100 ms at LEO RTT) completes before the handover fires, allowing BBR’s ProbeBW_UP to re-inflate the cwnd. Operators can treat $\ell \in [0, 2]$ s as the effective signal window for this transition.

A. Limitations and Future Work

Orbit table staleness. The 10-second `min_rtt` expiry causes BBR-SAT to fail for handover times $T_{HO} \geq 120$ s. Refreshing the orbit table entry during the PREDICTED window — a one-line change in the PREDICTED handler — would fix this at the cost of a brief ProbeBW_DOWN that re-seeds `min_rtt`.

Loss injection. All experiments are zero-loss. Satellite links exhibit correlated loss (rain fade, co-channel interference) that may interact with BBRv3’s loss-recovery path. The interaction between the DRAIN phase and `is_in_recovery` state — currently cleared on PREDICTED — requires evaluation under realistic loss models.

Multi-path QUIC. Satellite multi-path scenarios (e.g., LEO primary + GEO backup with simultaneous active paths) involve per-path BDP state and cross-path pacing interaction not addressed here.

BBR/CUBIC coexistence. The 55/45 CUBIC advantage in steady-state GEO (F3, $J = 0.887$) and the 86/14 split during F1-CUBIC ($J \approx 0.69$) are inherited from vanilla BBRv3’s loss-averse behaviour on large-BDP links. Addressing this would require either ECN-based queue-occupancy signalling or a fairness-aware pacing mode in BBRv3 itself.

VII. CONCLUSION

We presented BBR-SAT, a minimal extension of BBRv3 that equips QUIC with orbit-aware congestion control for satellite inter-orbit handovers. By integrating an ephemeris-seeded orbit parameter table and a two-phase PREDICTED/CONFIRMED signal protocol, BBR-SAT eliminates the $8 \times \text{RTT}$ stall inherent in BBRv3’s bandwidth filter on downward BDP transitions. In a zero-loss simulator spanning all six pairwise LEO/MEO/GEO transitions under a realistic $1 \times \text{BDP}$ droptail buffer cap, BBR-SAT is the only evaluated algorithm to converge consistently on the critical LEO→GEO transition, achieving $T_{90} = 4.5$ s at zero lead time (1.5 s with a 2-second advance signal) with 99 % link utilisation and 46 % more goodput than vanilla BBRv3 (20.1 MB vs. 13.7 MB). CUBIC converges in 5.5 s via packet loss; all other BBR baselines fail to converge. In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by $44 \times$ over vanilla BBRv3. Fairness analysis confirms that advance knowledge does not translate into a bandwidth advantage: BBR-SAT co-exists equitably with uninformed BBRv3 flows ($J = 0.994$) and preserves the existing BBRv3/CUBIC balance.

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